PromoIQ — AI-Powered Microdrama Promo Discovery System

Candidate Solution Document | Kuku AI Generalist Assignment — **Problem 1 (Find the Promos)** | Series: *I Married a Broken Soldier. He was a Duke*

1. Problem Framing, Business Context & Core Assumptions

The Opportunity: Microdrama series live or die on user acquisition funnels across TikTok, Instagram Reels, and YouTube Shorts. Manual promo selection by editors is slow, subjective, and creates release bottlenecks. PromoIQ automates the discovery of 20–60s clips that maximize viewer conversion by enforcing three non-negotiables: a **3-second arresting hook**, a **cut-off cliffhanger**, and **standalone clarity**.
Key Assumptions: (1) *Pure Screenplay Input:* No video renders exist; cinematic hooks (slaps, tears, crash zooms) must be extracted from **ACTION:** and **CAMERA:** tags, while emotional stakes come from dialogue and emotion markers. (2) *Strict 20–60s Duration:* Over-length clips drop retention on Reels; under-20s clips lack narrative cohesion. (3) *Zero Mandatory Paid APIs:* The system runs 100% locally with zero external API fees, guaranteeing repeatable evaluation on new screenplays during interviews.

2. Technical Architecture & Hybrid AI Implementation

PromoIQ adopts a **modular hybrid architecture** combining deterministic rules for precision with semantic NLP for tension reasoning:
• **State-Machine Screenplay Parser (src/parser.py):** Handles varying episode formatting, isolates scene boundaries, and normalizes (MM:SS.ms) timestamps into millisecond-accurate floats, enforcing strict monotonicity ($T_end \geq T_start$).
• **Semantic AI Analysis Engines (src/analysis.py):** Evaluates four independent dimensions: (A) *Hook Detector (0–3.5s):* Scans for high-arousal speech acts, shock vocabulary (betrayal, altar, poison), and kinetic actions. (B) *Cliffhanger Detector:* Assesses cut-off suspense via unanswered questions (?), interrupted speech (...), and ultimatums, severely penalizing polite conversational closures. (C) *Standalone Clarity:* Penalizes opening with dangling pronouns (*'He told me that...'*) and rewards multi-speaker interpersonal conflict. (D) *Emotional Engagement:* Tracks tension escalation and emotion tags (*Cold, Desperate, Sneering*).
• **Sliding-Window Candidate Generator & 1D NMS (src/generator.py):** Evaluates continuous 20–60s spans, intra-episode montages (Hook + Climax cut), and cross-episode bridges. Employs **1D Temporal Non-Maximum Suppression (IoU $\geq$ 0.60)** to discard redundant overlapping windows and retain only the single strongest variation.

3. Explainable Scoring Framework & Attribution Formula

Candidates receive a normalized composite score out of 100: **Score = 0.30 × S_hook + 0.30 × S_cliff + 0.20 × S_clarity + 0.15 × S_emotion + 0.05 × S_continuity**
Marketing Retention Note: A 98/100 score indicates high creative tension in the screenplay text. However, true viral CTR is influenced by non-textual elements (actor delivery, thumbnail frame, audio mix). Algorithmic scores prioritize candidates for editors, but final creative validation requires live A/B testing on Meta and TikTok Ads.

Top Discovered Promotional Clips across 50 Episodes

Rank	Promo ID	Ep	Timestamps	Dur	Score	Key Dramatic Hook & Cliffhanger
#1	PROMO_EP011_MONTAGE	11	00:14.30 - 01:24.00	40.3s	**99.3**	Opens on high-tension confrontation; cuts abruptly on an unresolved family secret reveal.
#2	PROMO_EP036_1552	36	01:43.50 - 02:28.00	44.5s	**99.0**	Shock accusation of betrayal; cuts on unanswered interrogative cliffhanger ('?').
#3	PROMO_EP036_2179	36	02:23.20 - 03:12.20	49.0s	**97.6**	Intense close-up freeze-frame confrontation; ends on high-stakes ultimatum.
#4	PROMO_EP032_1080	32	00:47.80 - 01:19.40	31.6s	**97.5**	Immediate betrayal confrontation; sudden verbal escalation with zero resolution.
#5	PROMO_EP002_MONTAGE	2	00:00.00 - 01:19.00	37.5s	**97.2**	Montage: Juliana abandoned in storm → sudden arrival of wounded Duke's carriage.

4. Quality Assurance, Failure Modes & Benchmark Calibration

Auditing the Candidate Pool: Out of 974 raw sliding windows generated across all 50 episodes, our automated Quality Checker flagged **361 candidates (37.1%)** with clear marketing defects, leaving **613 clean verified promos (62.9% pass rate)**. Duration violations were 0 (100% compliant) and timestamp monotonicity errors were 0.
Real Examples of AI Bad Selections & Diagnostics:
• *Missing Cliffhanger (Anti-Climactic Resolution - 249 flags):* E.g., PROMO_EP001_840 (01:05 - 01:42). The scene opened with dramatic rejection, but ended after Juliana whispered *'Yes, father'* and quietly left. Viewers scrolling TikTok have no reason to click Episode 2 when tension has dissipated. Caught by penalizing polite closure tokens by -30 points.
• *Weak Opening Hooks (185 flags):* Windows starting during slow camera establishers before characters speak or act. Flagged when 0–3s arousal score fell below 45/100.
• *Dangling Pronoun Context (22 flags):* E.g., PROMO_EP004_210 opening with *'He told me that she had no choice'*. New viewers swipe away because 'he' and 'she' are unknown. Caught via early pronoun detection without proper noun grounding (-22 pts).
Ground-Truth Benchmark Alignment: Evaluated against manually reviewed human benchmarks on Episodes 1–5: Ep 1 Altar Betrayal matched at 00:11.30-00:48.00 (Score: 95.0, exact match); Ep 2 Rainstorm Abandonment matched at 00:00.00-01:19.00 (Score: 97.2, creative montage); Ep 3 Contract Marriage matched at 00:15.00-01:22.00 (Score: 94.8, aligned).

5. Scalability, Cost Analysis & Record of AI Tools

• **Production Scalability & Zero Cost:** The pipeline processes all 50 episodes in **7.8 seconds** locally on standard CPU hardware. By maintaining an offline deterministic NLP engine with pluggable LLM interfaces (BaseLLMProvider supporting Gemini/Groq), the marginal cost per series is **$0.00**, allowing instant scaling to hundreds of weekly releases.
• **AI Reflection (Mistake & Correction):** During initial parser development, the AI assistant assumed header colons were placed outside the asterisks (**CAMERA**:). Actual screenplay files had colons inside (**CAMERA:** and **MATRIARCH (Cold, Commanding):**), causing the initial unit test to fail with 0 units. We resolved this by updating the regex to r'^**([^*]+?):?**:?\s* ', stripping redundant colons, and adding automated regression assertions in tests/test_parser.py.

Submission Deliverables Summary

Item	File / Artifact	Key Highlights & Verification
1. Solution Doc	outputs/Solution_Doc_PromoIQ_Kuku.pdf	2-Page concise PDF detailing architecture, scoring, quality failure audit, and AI reflection.
2. Prototype App	app.py (Streamlit)	Interactive 7-tab UI with 9:16 mobile reel cards, script deep-dive, dynamic re-scoring, and exports.
3. Raw Outputs	outputs/promos_discovered.csv & .json	Complete unedited dataset outputs (974 candidates with timestamps, sub-scores, and explanations).
4. Quality Report	outputs/quality_evaluation_report.md	Exhaustive audit breakdown, ground-truth calibration table, and diagnostic failure examples.
5. Codebase & Tests	src/, tests/, README.md	Clean modular architecture; 5/5 automated unit tests passing (100% free and runnable).